Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(Empowered Autonomous Institution under R.T.M Nagpur University)
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NAAC A++ Grade
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Department of CSE-IoT

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering: Dynamic learning environment that encourages problem-solving using emerging technology, · National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry–institute collaboration, Holistic, ethical and sustainable development for lifelong success.

Session 2025-2026(EVEN)

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Sheikh Wasimuddin
Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	2024-25 (EVEN)	Course Name	Competitive Coding
Semester	6	Course Code	23IOT1605
Roll No	63	Name of Student	Sheikh Wasimuddin
Practical Number	Competitive Coding DS Slot - 1		

Advanced Data Structure Problems in C

1. **Design and implement a Least Recently Used (LRU) Cache with capacity C.**

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB DS
#include <stdio.h>
#include <stdlib.h>
struct Node
{
   int key, value;
   struct Node *prev, *next;
};
struct Node *head = NULL, *tail = NULL;
struct Node *cache[1000];
int capacity, size = 0;
struct Node* createNode(int key, int value)
{
   struct Node* node = (struct Node*)malloc(sizeof(struct Node));
   node->key = key;
   node->value = value;
   node->prev = node->next = NULL;
   return node;
}
void moveToFront(struct Node* node)
{
   if (node == head)
      return;
   if (node->prev)
      node->prev->next = node->next;
   if (node->next)
      node->next->prev = node->prev;
   if (node == tail)
      tail = node->prev;
   node->prev = NULL;
   node->next = head;
```

```
    if (head)
        head->prev = node;
    head = node;
    if (!tail)
        tail = node;
}
void insertFront(struct Node* node)
{
    node->next = head;
    node->prev = NULL;
    if (head)
        head->prev = node;
    head = node;
    if (!tail)
        tail = node;
}
int get(int key)
{
    if (cache[key] == NULL)
        return -1;
    moveToFront(cache[key]);
    return cache[key]->value;
}
void put(int key, int value)
{
    if (cache[key] != NULL)
    {
        cache[key]->value = value;
        moveToFront(cache[key]);
        return;
    }
    struct Node* node = createNode(key, value);
    if (size == capacity)
    {
        cache[tail->key] = NULL;
        struct Node* temp = tail;
        tail = tail->prev;
        if (tail)
            tail->next = NULL;
        free(temp);
        size--;
    }
    insertFront(node);
    cache[key] = node;
    size++;
}
```

```
int main()
{
    int n, type, key, value;
    printf("Enter capacity: ");
    scanf("%d", &capacity);
    for (int i = 0; i < 1000; i++)
        cache[i] = NULL;
    printf("Enter number of operations: ");
    scanf("%d", &n);
    while (n--)
    {
        printf("Enter type (1-put, 2-get): ");
        scanf("%d", &type);
        if (type == 1)
        {
            printf("Enter key and value: ");
            scanf("%d %d", &key, &value);
            put(key, value);
        }
        else
        {
            printf("Enter key: ");
            scanf("%d", &key);
            int res = get(key);
            if (res == -1)
                printf("Not found\n");
            else
                printf("Value: %d\n", res);
        }
    }
    return 0;
}
```

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main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB DS
#include <stdio.h>
#include <stdlib.h>
struct Node
{
int key, value;
struct Node *prev, *next;
};
struct Node *head = NULL, *tail = NULL; struct Node *cache[1000];
```

input

```
Enter capacity: 2
Enter number of operations: 5
Enter type (1-put, 2-get): 1
Enter key and value: 1 10
Enter type (1-put, 2-get): 1
Enter key and value: 2 20
Enter type (1-put, 2-get): 2
Enter key: 1
Value: 10
Enter type (1-put, 2-get): 1
Enter key and value: 3 30
Enter type (1-put, 2-get): 2
Enter key: 2
Not found
```

2. **Implement a Trie that supports inserting words and searching with wildcards. The . character can match any single character.**

Answer)

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
struct Trie
{
   struct Trie *child[26];
   int end;
};
struct Trie* createNode()
{
   struct Trie *node = (struct Trie*)malloc(sizeof(struct Trie));
   for (int i = 0; i < 26; i++)
      node->child[i] = NULL;
   node->end = 0;
   return node;
}

void insert(struct Trie *root, char *word)
{
```

```
    struct Trie *cur = root;
    for (int i = 0; word[i]; i++)
    {
        int idx = word[i] - 'a';
        if (!cur->child[idx])
            cur->child[idx] = createNode();
        cur = cur->child[idx];
    }
    cur->end = 1;
}

int search(struct Trie *root, char *word, int pos)
{
    if (!root)
        return 0;

    if (word[pos] == '\0')
        return root->end;

    if (word[pos] == '.')
    {
        for (int i = 0; i < 26; i++)
        {
            if (search(root->child[i], word, pos + 1))
                return 1;
        }
        return 0;
    }
    else
    {
        int idx = word[pos] - 'a';
        return search(root->child[idx], word, pos + 1);
    }
}

int main()
{
    struct Trie *root = createNode();
    int n;
    char word[50];

    printf("Enter number of words: ");
    scanf("%d", &n);

    printf("Enter words:\n");
    for (int i = 0; i < n; i++)
```

```
    {
        scanf("%s", word);
        insert(root, word);
    }

    printf("Enter search pattern: ");
    scanf("%s", word);

    if (search(root, word, 0))
        printf("Word found");
    else
        printf("Word not found");

    return 0;
}
```

main.c

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
struct Trie
{
struct Trie *child[26]; int end;
};
struct Trie* createNode()
{
struct Trie *node = (struct Trie*)malloc(sizeof(struct Trie)); for (int i = 0; i < 26; i++)
```

input

```
Enter number of words: 3
Enter words:
bat
mat
rat
Enter search pattern: ..t
Word found

...Program finished with exit code 0
Press ENTER to exit console.
```

3. **Implement a circular buffer (ring buffer) of size N supporting enqueue and dequeue operations.**

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
int queue[100];
int front = -1, rear = -1, size;

int isFull()
{
   return (front == (rear + 1) % size);
}

int isEmpty()
{
   return (front == -1);
}

void enqueue(int value)
{
   if (isFull())
   {
      printf("Buffer Full\n");
      return;
   }

   if (front == -1)
      front = 0;

   rear = (rear + 1) % size;
   queue[rear] = value;
}

void dequeue()
{
   if (isEmpty())
   {
      printf("Buffer Empty\n");
      return;
   }

   printf("Removed: %d\n", queue[front]);
```

```
    if (front == rear)
    {
        front = rear = -1;
    }
    else
    {
        front = (front + 1) % size;
    }
}

void display()
{
    if (isEmpty())
    {
        printf("Buffer Empty\n");
        return;
    }

    int i = front;
    while (1)
    {
        printf("%d ", queue[i]);
        if (i == rear)
            break;
        i = (i + 1) % size;
    }
    printf("\n");
}

int main()
{
    int n, choice, value;

    printf("Enter buffer size: ");
    scanf("%d", &size);

    printf("Enter number of operations: ");
    scanf("%d", &n);

    while (n--)
    {
        printf("1-Enqueue 2-Dequeue 3-Display: ");
        scanf("%d", &choice);

        if (choice == 1)
        {
```

```
            printf("Enter value: ");
            scanf("%d", &value);
            enqueue(value);
        }
        else if (choice == 2)
        {
            dequeue();
        }
        else
        {
            display();
        }
    }

    return 0;
}
```

main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
int queue[100];
int front = -1, rear = -1, size;

int isFull()
{
return (front == (rear + 1) % size);
}

```

input

```
Enter buffer size: 3
Enter number of operations: 5
1-Enqueue 2-Dequeue 3-Display: 1
Enter value: 30
1-Enqueue 2-Dequeue 3-Display: 1
Enter value: 20
1-Enqueue 2-Dequeue 3-Display: 1
Enter value: 10
1-Enqueue 2-Dequeue 3-Display: 2
Removed: 30
1-Enqueue 2-Dequeue 3-Display: 3
20 10
```

4. **Implement an Interval Tree that stores intervals [low, high] and supports efficient queries to find all intervals overlapping with a given interval.**

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
struct Node
{
   int low, high, max;
   struct Node *left, *right;
};
struct Node* createNode(int low, int high)
{
   struct Node* node = (struct Node*)malloc(sizeof(struct Node));
   node->low = low;
   node->high = high;
   node->max = high;
   node->left = node->right = NULL;
   return node;
}
int max(int a, int b)
{
   return a > b ? a : b;
}
struct Node* insert(struct Node* root, int low, int high)
{
   if (!root)
      return createNode(low, high);

   if (low < root->low)
      root->left = insert(root->left, low, high);
   else
      root->right = insert(root->right, low, high);

   if (root->max < high)
      root->max = high;

   return root;
}

int overlap(int l1, int h1, int l2, int h2)
{
   return (l1 <= h2 && l2 <= h1);
```

```
}

void searchOverlap(struct Node* root, int low, int high)
{
   if (!root)
      return;

   if (overlap(root->low, root->high, low, high))
      printf("[%d,%d] ", root->low, root->high);

   if (root->left && root->left->max >= low)
      searchOverlap(root->left, low, high);

   searchOverlap(root->right, low, high);
}

int main()
{
   struct Node* root = NULL;
   int n, i, low, high;

   printf("Enter number of intervals: ");
   scanf("%d", &n);

   printf("Enter intervals (low high):\n");
   for (i = 0; i < n; i++)
   {
      scanf("%d %d", &low, &high);
      root = insert(root, low, high);
   }

   printf("Enter query interval: ");
   scanf("%d %d", &low, &high);

   printf("Overlapping intervals: ");
   searchOverlap(root, low, high);

   return 0;
}
```

main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
struct Node
{
int low, high, max; struct Node *left, *right;
};
struct Node* createNode(int low, int high)
{
struct Node* node = (struct Node*)malloc(sizeof(struct Node)); node->low = low;
```

input

```
Enter number of intervals: 4
Enter intervals (low high):
20 30
10 20
25 35
95 105
Enter query interval: 19 26
Overlapping intervals: [20,30] [10,20] [25,35]
```

5. **Implement a thread-safe queue that supports concurrent enqueue and dequeue operations using mutexes and condition variables.**

Answer)

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
#include <windows.h>
int queue[100];
int front = 0, rear = 0, count = 0, capacity;
CRITICAL_SECTION lock;
HANDLE not_full, not_empty;
void enqueue(int value)
{
   EnterCriticalSection(&lock);

   while (count == capacity)
   {
      LeaveCriticalSection(&lock);
      WaitForSingleObject(not_full, INFINITE);
      EnterCriticalSection(&lock);
   }
   queue[rear] = value;
   rear = (rear + 1) % capacity;
   count++;
```

```
    SetEvent(not_empty);
    LeaveCriticalSection(&lock);
}
int dequeue()
{
    EnterCriticalSection(&lock);

    while (count == 0)
    {
        LeaveCriticalSection(&lock);
        WaitForSingleObject(not_empty, INFINITE);
        EnterCriticalSection(&lock);
    }

    int value = queue[front];
    front = (front + 1) % capacity;
    count--;

    SetEvent(not_full);
    LeaveCriticalSection(&lock);

    return value;
}

DWORD WINAPI producer(LPVOID arg)
{
    int n = *(int*)arg;
    for (int i = 1; i <= n; i++)
    {
        enqueue(i);
        printf("Produced: %d\n", i);
    }
    return 0;
}

DWORD WINAPI consumer(LPVOID arg)
{
    int n = *(int*)arg;
    for (int i = 1; i <= n; i++)
    {
        int val = dequeue();
        printf("Consumed: %d\n", val);
    }
    return 0;
}
```

```
int main()
{
    int n;

    printf("Enter queue capacity: ");
    scanf("%d", &capacity);

    printf("Enter number of items: ");
    scanf("%d", &n);

    InitializeCriticalSection(&lock);

    not_full = CreateEvent(NULL, FALSE, TRUE, NULL);
    not_empty = CreateEvent(NULL, FALSE, FALSE, NULL);

    HANDLE t1 = CreateThread(NULL, 0, producer, &n, 0, NULL);
    HANDLE t2 = CreateThread(NULL, 0, consumer, &n, 0, NULL);

    WaitForSingleObject(t1, INFINITE);
    WaitForSingleObject(t2,  INFINITE);

    DeleteCriticalSection(&lock);
    CloseHandle(not_full);
    CloseHandle(not_empty);
```

```
#include <stdio.h>
#include <stdlib.h>
#include <windows.h>
int queue[100];
int front = 0, rear = 0, count = 0, capacity;
CRITICAL_SECTION lock;
HANDLE not_full, not_empty;
void enqueue(int value)
{
    EnterCriticalSection(&lock);
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\hp\OneDrive\Desktop\Coding\Advance Ds> cd "c:\Users\hp\OneDrive\Desktop\Coding\Advance Ds\" ; if ($?) { gcc Q5_thread-safe.c -o Q5_thre
Enter queue capacity: 3
Enter number of items: 5
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
Consumed: 5
Produced: 5
PS C:\Users\hp\OneDrive\Desktop\Coding\Advance Ds>
```

6. **Implement a Skip List that supports efficient insertion, deletion, and search operations.**

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#define MAX 5
struct Node
{
   int key;
   struct Node* forward[MAX];
};
struct Node* header;
int level = 0;
struct Node* createNode(int key)
{
   struct Node* node = (struct Node*)malloc(sizeof(struct Node));
   node->key = key;
   for (int i = 0; i < MAX; i++)
      node->forward[i] = NULL;
```

```
    return node;
}
int randomLevel()
{
    int lvl = 0;
    while (rand() % 2 && lvl < MAX - 1)
        lvl++;
    return lvl;
}
void insert(int key)
{
    struct Node* update[MAX];
    struct Node* x = header;

    for (int i = level; i >= 0; i--)
    {
        while (x->forward[i] && x->forward[i]->key < key)
            x = x->forward[i];
        update[i] = x;
    }

    x = x->forward[0];

    if (x == NULL || x->key != key)
    {
        int lvl = randomLevel();

        if (lvl > level)
        {
            for (int i = level + 1; i <= lvl; i++)
                update[i] = header;
            level = lvl;
        }

        x = createNode(key);

        for (int i = 0; i <= lvl; i++)
        {
            x->forward[i] = update[i]->forward[i];
            update[i]->forward[i] = x;
        }
    }
}

void deleteKey(int key)
{
```

```
    struct Node* update[MAX];
    struct Node* x = header;

    for (int i = level; i >= 0; i--)
    {
        while (x->forward[i] && x->forward[i]->key < key)
            x = x->forward[i];
        update[i] = x;
    }

    x = x->forward[0];

    if (x && x->key == key)
    {
        for (int i = 0; i <= level; i++)
        {
            if (update[i]->forward[i] != x)
                break;
            update[i]->forward[i] = x->forward[i];
        }

        free(x);

        while (level > 0 && header->forward[level] == NULL)
            level--;
    }
}

int search(int key)
{
    struct Node* x = header;

    for (int i = level; i >= 0; i--)
    {
        while (x->forward[i] && x->forward[i]->key < key)
            x = x->forward[i];
    }

    x = x->forward[0];

    if (x && x->key == key)
        return 1;
    return 0;
}

void display()
```

```
{
    struct Node* x = header->forward[0];
    while (x)
    {
        printf("%d ", x->key);
        x = x->forward[0];
    }
    printf("\n");
}

int main()
{
    int n, choice, key;

    srand(time(0));
    header = createNode(-1);

    printf("Enter number of operations: ");
    scanf("%d", &n);

    while (n--)
    {
        printf("1-Insert 2-Delete 3-Search 4-Display: ");
        scanf("%d", &choice);

        if (choice == 1)
        {
            printf("Enter key: ");
            scanf("%d", &key);
            insert(key);
        }
        else if (choice == 2)
        {
            printf("Enter key: ");
            scanf("%d", &key);
            deleteKey(key);
        }
        else if (choice == 3)
        {
            printf("Enter key: ");
            scanf("%d", &key);
            if (search(key))
                printf("Found\n");
            else
                printf("Not Found\n");
        }
```

```
        else
        {
            display();
        }
    }

    return 0;
}
```

main.c

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#define MAX 5
struct Node
{
int key;
struct Node* forward[MAX];
};
```

```
Enter number of operations: 5
1-Insert 2-Delete 3-Search 4-Display: 1
Enter key: 50
1-Insert 2-Delete 3-Search 4-Display: 1
Enter key: 30
1-Insert 2-Delete 3-Search 4-Display: 1
Enter key: 70
1-Insert 2-Delete 3-Search 4-Display: 4
30 50 70
1-Insert 2-Delete 3-Search 4-Display: 3
Enter key: 20
Not Found
```

7. **Implement a B-Tree of minimum degree t that supports insertion, deletion, and search operations.**

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
#define T 3
struct Node
{
    int keys[2*T-1];
    struct Node* child[2*T];
    int n;
    int leaf;
};
struct Node* createNode(int leaf)
```

```
{
    struct Node* node = (struct Node*)malloc(sizeof(struct Node));
    node->leaf = leaf;
    node->n = 0;
    for (int i = 0; i < 2*T; i++)
        node->child[i] = NULL;
    return node;
}
void traverse(struct Node* root)
{
    int i;
    for (i = 0; i < root->n; i++)
    {
        if (!root->leaf)
            traverse(root->child[i]);
        printf("%d ", root->keys[i]);
    }
    if (!root->leaf)
        traverse(root->child[i]);
}

struct Node* search(struct Node* root, int k)
{
    int i = 0;
    while (i < root->n && k > root->keys[i])
        i++;

    if (i < root->n && root->keys[i] == k)
        return root;

    if (root->leaf)
        return NULL;

    return search(root->child[i], k);
}

void splitChild(struct Node* x, int i, struct Node* y)
{
    struct Node* z = createNode(y->leaf);
    z->n = T - 1;

    for (int j = 0; j < T - 1; j++)
        z->keys[j] = y->keys[j + T];

    if (!y->leaf)
    {
```

```
        for (int j = 0; j < T; j++)
            z->child[j] = y->child[j + T];
    }

    y->n = T - 1;

    for (int j = x->n; j >= i + 1; j--)
        x->child[j + 1] = x->child[j];

    x->child[i + 1] = z;

    for (int j = x->n - 1; j >= i; j--)
        x->keys[j + 1] = x->keys[j];

    x->keys[i] = y->keys[T - 1];
    x->n++;
}

void insertNonFull(struct Node* x, int k)
{
    int i = x->n - 1;

    if (x->leaf)
    {
        while (i >= 0 && x->keys[i] > k)
        {
            x->keys[i + 1] = x->keys[i];
            i--;
        }
        x->keys[i + 1] = k;
        x->n++;
    }
    else
    {
        while (i >= 0 && x->keys[i] > k)
            i--;

        if (x->child[i + 1]->n == 2*T - 1)
        {
            splitChild(x, i + 1, x->child[i + 1]);
            if (k > x->keys[i + 1])
                i++;
        }
        insertNonFull(x->child[i + 1], k);
    }
}
```

```
struct Node* insert(struct Node* root, int k)
{
   if (root->n == 2*T - 1)
   {
      struct Node* s = createNode(0);
      s->child[0] = root;
      splitChild(s, 0, root);

      int i = 0;
      if (s->keys[0] < k)
         i++;
      insertNonFull(s->child[i],  k);

      return s;
   }
   else
   {
      insertNonFull(root, k);
      return root;
   }
}

int main()
{
   struct Node* root = createNode(1);
   int n, choice, key;

   printf("Enter number of operations: ");
   scanf("%d", &n);

   while (n--)
   {
      printf("1-Insert 2-Search 3-Display: ");
      scanf("%d", &choice);

      if (choice == 1)
      {
         printf("Enter key: ");
         scanf("%d", &key);
         root = insert(root, key);
      }
      else if (choice == 2)
      {
         printf("Enter key: ");
         scanf("%d", &key);
```

```
            if (search(root, key))
                printf("Found\n");
            else
                printf("Not Found\n");
        }
        else
        {
            traverse(root);
            printf("\n");
        }
    }

    return 0;
}
```

main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
#define T 3
struct Node
{
int keys[2*T-1];
struct Node* child[2*T]; int n;
int leaf;
};
```

```
Enter number of operations: 5
1-Insert 2-Search 3-Display: 1
Enter key: 7
1-Insert 2-Search 3-Display: 1
Enter key: 10
1-Insert 2-Search 3-Display: 1
Enter key: 18
1-Insert 2-Search 3-Display: 3
7 10 18
1-Insert 2-Search 3-Display: 2
Enter key: 9
Not Found
```

8. Implement a Union-Find (Disjoint Set) data structure with path compression and union by rank.

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
int parent[100], rankArr[100];
void makeSet(int n)
{
   for(int i=0;i<n;i++)
   {
      parent[i]=i;
      rankArr[i]=0;
   }
}
int find(int x)
{
   if(parent[x]!=x)
      parent[x]=find(parent[x]);
   return parent[x];
}
void unionSet(int x,int y)
{
   int rootX=find(x);
   int rootY=find(y);
   if(rootX!=rootY)
   {
      if(rankArr[rootX]<rankArr[rootY])
         parent[rootX]=rootY;
      else if(rankArr[rootX]>rankArr[rootY])
         parent[rootY]=rootX;
      else
      {
         parent[rootY]=rootX;
         rankArr[rootX]++;
      }
   }
}

int main()
{
   int n,m;

   printf("Enter number of elements: ");
   scanf("%d",&n);
```

```
    makeSet(n);

    printf("Enter number of unions: ");
    scanf("%d",&m);

    for(int i=0;i<m;i++)
    {
        int x,y;
        printf("Enter pair %d: ",i+1);
        scanf("%d %d",&x,&y);
        unionSet(x,y);
    }

    printf("Set representatives:\n");
    for(int i=0;i<n;i++)
        printf("%d ",find(i));

    return 0;
}
```

main.c

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
int parent[100],rankArr[100];
void makeSet(int n)
{
for(int i=0;i<n;i++)
{
parent[i]=i; rankArr[i]=0;
}
}
```

```
Enter number of elements: 5
Enter number of unions: 3
Enter pair 1: 0 1
Enter pair 2: 1 2
Enter pair 3: 2 3
Set representatives:
0 0 0 0 4
```

9. **Given a weighted directed graph, implement Dijkstra's Algorithm using adjacency lists to find the shortest distance from a source vertex to all others.**

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#define MAX 100
#define INF 9999
int n, e;
int adj[MAX][MAX];
void dijkstra(int src)
{
   int dist[MAX], visited[MAX];

   for(int i=0;i<n;i++)
   {
      dist[i]=INF;
      visited[i]=0;
   }
   dist[src]=0;
   for(int i=0;i<n-1;i++)
   {
      int min=INF, u;
      for(int j=0;j<n;j++)
      {
         if(!visited[j] && dist[j]<min)
         {
            min=dist[j];
            u=j;
         }
      }
      visited[u]=1;
      for(int v=0;v<n;v++)
      {
         if(!visited[v] && adj[u][v]!=INF && dist[u]+adj[u][v]<dist[v])
            dist[v]=dist[u]+adj[u][v];
      }
   }

   printf("Shortest distances from source:\n");
   for(int i=0;i<n;i++)
      printf("To %d = %d\n",i,dist[i]);
}
```

```
int main()
{
    printf("Enter number of vertices: ");
    scanf("%d",&n);

    printf("Enter number of edges: ");
    scanf("%d",&e);

    for(int i=0;i<n;i++)
        for(int j=0;j<n;j++)
            adj[i][j]=INF;

    printf("Enter edges (u v weight):\n");
    for(int i=0;i<e;i++)
    {
        int u,v,w;
        scanf("%d %d %d",&u,&v,&w);
        adj[u][v]=w;
    }

    int src;
    printf("Enter source vertex: ");
    scanf("%d",&src);

    dijkstra(src);

    return 0;
}
```

main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#define MAX 100
#define INF 9999
int n, e;
int adj[MAX][MAX];
void dijkstra(int src)
{
int dist[MAX], visited[MAX];
```

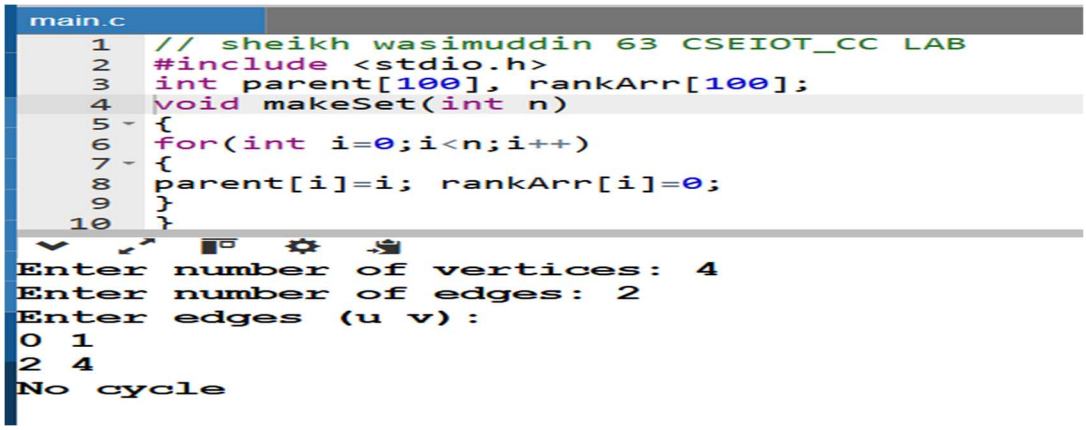
```
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (u v weight):
0 2 1
2 0 4
1 2 1
3 7 1
2 4 3
Enter source vertex: 1
Shortest distances from source:
To 0 = 5
To 1 = 0
To 2 = 1
To 3 = 9999
```

10. Detect whether a given undirected graph contains a cycle using Union-Find.

Answer)

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
int parent[100], rankArr[100];
void makeSet(int n)
{
   for(int i=0;i<n;i++)
   {
      parent[i]=i;
      rankArr[i]=0;
   }
}
int find(int x)
{
   if(parent[x]!=x)
      parent[x]=find(parent[x]);
   return parent[x];
}
int unionSet(int x,int y)
```

```
{
    int rootX=find(x);
    int rootY=find(y);

    if(rootX==rootY)
        return 1;
    if(rankArr[rootX]<rankArr[rootY])
        parent[rootX]=rootY;
    else if(rankArr[rootX]>rankArr[rootY])
        parent[rootY]=rootX;
    else
    {
        parent[rootY]=rootX;
        rankArr[rootX]++;
    }
    return 0;
}
int main()
{
    int n,e;
    printf("Enter number of vertices: ");
    scanf("%d",&n);

    printf("Enter number of edges: ");
    scanf("%d",&e);

    makeSet(n);

    int cycle=0;

    printf("Enter edges (u v):\n");
    for(int i=0;i<e;i++)
    {
        int u,v;
        scanf("%d %d",&u,&v);

        if(unionSet(u,v))
            cycle=1;
    }

    if(cycle)
        printf("Cycle detected\n");
    else
        printf("No cycle\n");

    return 0; }
```



11. Implement a memory-efficient representation of a sparse matrix.

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
int main()
{
   int rows, cols;
   printf("Enter rows and columns: ");
   scanf("%d %d",&rows,&cols);
   int matrix[100][100];
   printf("Enter matrix elements:\n");
   for(int i=0;i<rows;i++)
      for(int j=0;j<cols;j++)
         scanf("%d",&matrix[i][j]);
   int sparse[100][3];
   int k=0;
   for(int i=0;i<rows;i++)
   {
      for(int j=0;j<cols;j++)
      {
```

```
            if(matrix[i][j]!=0)
            {
                sparse[k][0]=i;
                sparse[k][1]=j;
                sparse[k][2]=matrix[i][j];
                k++;
            }
        }
    }
    printf("Sparse Matrix (row col value):\n");
    for(int i=0;i<k;i++)
        printf("%d %d %d\n",sparse[i][0],sparse[i][1],sparse[i][2]);
    return 0;
}
```

main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
int main()
{
int rows, cols;
printf("Enter rows and columns: "); scanf("%d %d",&rows,&cols);
int matrix[100][100]; printf("Enter matrix elements:\n"); for(int i=0;i<rows;i++)
for(int j=0;j<cols;j++) scanf("%d",&matrix[i][j]);
int sparse[100][3]; int k=0;
for(int i=0;i<rows;i++)
```

input

```
Enter rows and columns: 3 3
Enter matrix elements:
1 0 1
0 1 0
1 0 1
Sparse Matrix (row col value):
0 0 1
0 2 1
1 1 1
2 0 1
2 2 1
```

12. Implement a Bloom Filter for strings with multiple hash functions.

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <string.h>
#define SIZE 100
int bitArray[SIZE];
int hash1(char *str)
```

```
{
    int hash=0;

    for(int i=0;str[i];i++)
        hash=(hash+str[i])%SIZE;
    return hash;
}
int hash2(char *str)
{
    int hash=0;
    for(int i=0;str[i];i++)
        hash=(hash*31+str[i])%SIZE;
    return hash;
}
void insert(char *str)
{
    int h1=hash1(str);
    int h2=hash2(str);

    bitArray[h1]=1;
    bitArray[h2]=1;
}
int check(char *str)
{
    int h1=hash1(str);
    int h2=hash2(str);

    if(bitArray[h1] && bitArray[h2])
        return 1;
    return 0;
}
int main()
{
    int n;
    char str[50];

    printf("Enter number of strings to insert: ");
    scanf("%d",&n);

    for(int i=0;i<n;i++)
    {
        printf("Enter string %d: ",i+1);
        scanf("%s",str);
        insert(str);
    }

    int q;
    printf("Enter number of queries: ");
```

```
    scanf("%d",&q);


    for(int i=0;i<q;i++)
    {
        printf("Enter string to check: ");
        scanf("%s",str);

        if(check(str))
            printf("Possibly present\n");
        else
            printf("Definitely not present\n");
    }

    return 0;
}
```

main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <string.h>
#define SIZE 100
int bitArray[SIZE];
int hash1(char *str)
{
int hash=0;

for(int i=0;str[i];i++) hash=(hash+str[i])%SI
```

```
Enter number of strings to insert: 4
Enter string 1: cat
Enter string 2: bat
Enter string 3: mat
Enter string 4: rat
Enter number of queries: 4
Enter string to check: bat
Possibly present
Enter string to check: pat
Definitely not present
Enter string to check: rat
Possibly present
Enter string to check: net
Definitely not present
```

13. Implement a Segment Tree supporting: 1) Range sum queries 2) Range add updates

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#define MAX 100
```

```
int arr[MAX], tree[4*MAX], lazy[4*MAX];
void build(int node,int start,int end)
{
   if(start==end)
      tree[node]=arr[start];

   else
   {
      int mid=(start+end)/2;
      build(2*node,start,mid);
      build(2*node+1,mid+1,end);
      tree[node]=tree[2*node]+tree[2*node+1];
   }
}
void update(int node,int start,int end,int l,int r,int val)
{
   if(lazy[node]!=0)
   {
      tree[node]+=(end-start+1)*lazy[node];
      if(start!=end)
      {
         lazy[2*node]+=lazy[node];
         lazy[2*node+1]+=lazy[node];
      }
      lazy[node]=0;
   }
   if(start>r || end<l)
      return;
   if(start>=l && end<=r)
   {
      tree[node]+=(end-start+1)*val;
      if(start!=end)
      {
         lazy[2*node]+=val;
         lazy[2*node+1]+=val;
      }
      return;
   }
   int mid=(start+end)/2;
   update(2*node,start,mid,l,r,val);
   update(2*node+1,mid+1,end,l,r,val);

   tree[node]=tree[2*node]+tree[2*node+1];
}
int query(int node,int start,int end,int l,int r)
{
   if(start>r || end<l)
      return 0;
```

```
    if(lazy[node]!=0)
    {
        tree[node]+=(end-start+1)*lazy[node];

        if(start!=end)
        {
            lazy[2*node]+=lazy[node];
            lazy[2*node+1]+=lazy[node];
        }
        lazy[node]=0;
    }
    if(start>=l && end<=r)
        return tree[node];

    int mid=(start+end)/2;
    return query(2*node,start,mid,l,r) + query(2*node+1,mid+1,end,l,r);
}
int main()
{
    int n;
    printf("Enter number of elements: ");
    scanf("%d",&n);
    printf("Enter elements:\n");
    for(int i=0;i<n;i++)
        scanf("%d",&arr[i]);
    build(1,0,n-1);
    int q;
    printf("Enter number of operations: ");
    scanf("%d",&q);
    for(int i=0;i<q;i++)
    {
        int type;
        printf("1: Update  2: Query\nEnter type: ");
        scanf("%d",&type);

        if(type==1)
        {
            int l,r,val;
            printf("Enter l r value: ");
            scanf("%d %d %d",&l,&r,&val);
            update(1,0,n-1,l,r,val);
        }
        else
        {
            int l,r;
            printf("Enter l r: ");
            scanf("%d %d",&l,&r);
```

```
        printf("Sum = %d\n",query(1,0,n-1,l,r));
      }
    }


    return 0;
  }
```

```
main.c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#define MAX 100
int arr[MAX], tree[4*MAX], lazy[4*MAX];
void build(int node,int start,int end)
{
if(start==end) tree[node]=arr[start];

else
```

```
Enter number of elements: 5
Enter elements:
9 8 7 6 5
Enter number of operations: 5
1: Update 2: Query
Enter type: 1
Enter l r value: 1 2 3 4 5
1: Update 2: Query
Enter type: Enter l r: 2
Sum = 0
1: Update 2: Query
Enter type: 2
Enter l r: 0 4
Sum = 41
1: Update 2: Query
Enter type: 1
Enter l r value: 2 3 5
```

14. Given an array, count the number of inversions (i, j) where i<t; j and arr[i] > arr[j] using Binary Indexed Tree (Fenwick Tree).

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#define MAX 100
int BIT[MAX], n;
void update(int i,int val)
{
  while(i<=n)
  {
    BIT[i]+=val;
    i+=i&(-i);
  }
}
```

```
int query(int i)
{
    int sum=0;
    while(i>0)
    {
        sum+=BIT[i];
        i-=i&(-i);
    }
    return sum;
}
int main()
{
    int arr[MAX], temp[MAX];

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(int i=1;i<=n;i++)
    {
        scanf("%d",&arr[i]);
        temp[i]=arr[i];
    }
    for(int i=1;i<=n;i++)
    {
        for(int j=i+1;j<=n;j++)
        {
            if(temp[i]>temp[j])
            {
                int t=temp[i];
                temp[i]=temp[j];
                temp[j]=t;
            }
        }
    }
    for(int i=1;i<=n;i++)
    {
        for(int j=1;j<=n;j++)
        {
            if(arr[i]==temp[j])
            {
                arr[i]=j;
                break;
            }
        }
    }
    int inv=0;
    for(int i=n;i>=1;i--)
```

```
    {
        inv+=query(arr[i]-1);
        update(arr[i],1);

    }
    printf("Number of inversions = %d\n",inv);
    return 0;
}
```

main.c

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#define MAX 100
int BIT[MAX], n;
void update(int i,int val)
{
while(i<=n)
{
BIT[i]+=val; i+=i&(-i);
}
```

```
Enter number of elements: 6
Enter elements:
9 8 7 6 5 4
Number of inversions = 15
```

15. Design a data structure Fibonacci Heap in C that supports efficient priority queue operations.

Answer)

```
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
typedef struct node
{
    int key, degree;
    struct node *parent, *child, *left, *right;
}node;
node *min=NULL;
node* createNode(int key)
{
    node* newNode=(node*)malloc(sizeof(node));
    newNode->key=key;
    newNode->degree=0;
    newNode->parent=newNode->child=NULL;
    newNode->left=newNode->right=newNode;
    return newNode;
```

```
}
void insert(int key)
{
   node* x=createNode(key);
   if(min==NULL)
      min=x;
   else
   {
      x->left=min;
      x->right=min->right;
      min->right->left=x;
      min->right=x;

      if(x->key<min->key)
         min=x;
   }
}
node* getMin()
{
   return min;
}
void link(node* y,node* x)
{
   y->left->right=y->right;
   y->right->left=y->left;
   y->parent=x;
   if(x->child==NULL)
   {
      x->child=y;
      y->left=y->right=y;
   }
   else
   {
      y->left=x->child;
      y->right=x->child->right;
      x->child->right->left=y;
      x->child->right=y;
   }

   x->degree++;
}
void consolidate()
{
   node* A[100]={0};
   node* w=min;
   if(w==NULL) return;
```

```
    node* start=w;
    do
    {
        node* x=w;
        int d=x->degree;
        while(A[d]!=NULL)
        {
            node* y=A[d];
            if(x->key>y->key)
            {
                node* temp=x;
                x=y;
                y=temp;
            }
            link(y,x);
            A[d]=NULL;
            d++;
        }
        A[d]=x;
        w=w->right;
    }while(w!=start);
    min=NULL;
    for(int i=0;i<100;i++)
    {
        if(A[i]!=NULL)
        {
            if(min==NULL)
                min=A[i];
            else if(A[i]->key<min->key)
                min=A[i];
        }
    }
}
void extractMin()
{
    node* z=min;
    if(z!=NULL)
    {
        if(z->child!=NULL)
        {
            node* c=z->child;
            node* start=c;
            do
            {
                node* next=c->right;
                c->left=min;
```

```
            c->right=min->right;
            min->right->left=c;
            min->right=c;
            c->parent=NULL;
            c=next;
         }while(c!=start);
      }
      z->left->right=z->right;
      z->right->left=z->left;
      if(z==z->right)
         min=NULL;
      else
      {
         min=z->right;
         consolidate();
      }
   }
}
int main()
{
   int choice, val;
   while(1)
   {
      printf("\n1.Insert 2.GetMin 3.ExtractMin 4.Exit\n");
      printf("Enter choice: ");
      scanf("%d",&choice);
      if(choice==1)
      {
         printf("Enter value: ");
         scanf("%d",&val);
         insert(val);
      }
      else if(choice==2)
      {
         if(min)
            printf("Minimum = %d\n",getMin()->key);
         else
            printf("Heap is empty\n");
      }
      else if(choice==3)
      {
         if(min)
         {
            printf("Extracted Min = %d\n",min->key);
            extractMin();
         }
```

```
            else
                printf("Heap is empty\n");
        }
        else
            break;
    }
    return 0;
}
```

main.c

```c
// sheikh wasimuddin 63 CSEIOT_CC LAB
#include <stdio.h>
#include <stdlib.h>
typedef struct node
{
int key, degree;
struct node *parent, *child, *left, *right;
}node;
node *min=NULL;
node* createNode(int key)
```

```
1.Insert 2.GetMin 3.ExtractMin 4.Exit
Enter choice: 1
Enter value: 77

1.Insert 2.GetMin 3.ExtractMin 4.Exit
Enter choice: 1
Enter value: 99

1.Insert 2.GetMin 3.ExtractMin 4.Exit
Enter choice: 1
Enter value: 268

1.Insert 2.GetMin 3.ExtractMin 4.Exit
Enter choice: 3
Extracted Min = 77

1.Insert 2.GetMin 3.ExtractMin 4.Exit
Enter choice: 2
Minimum = 99
```